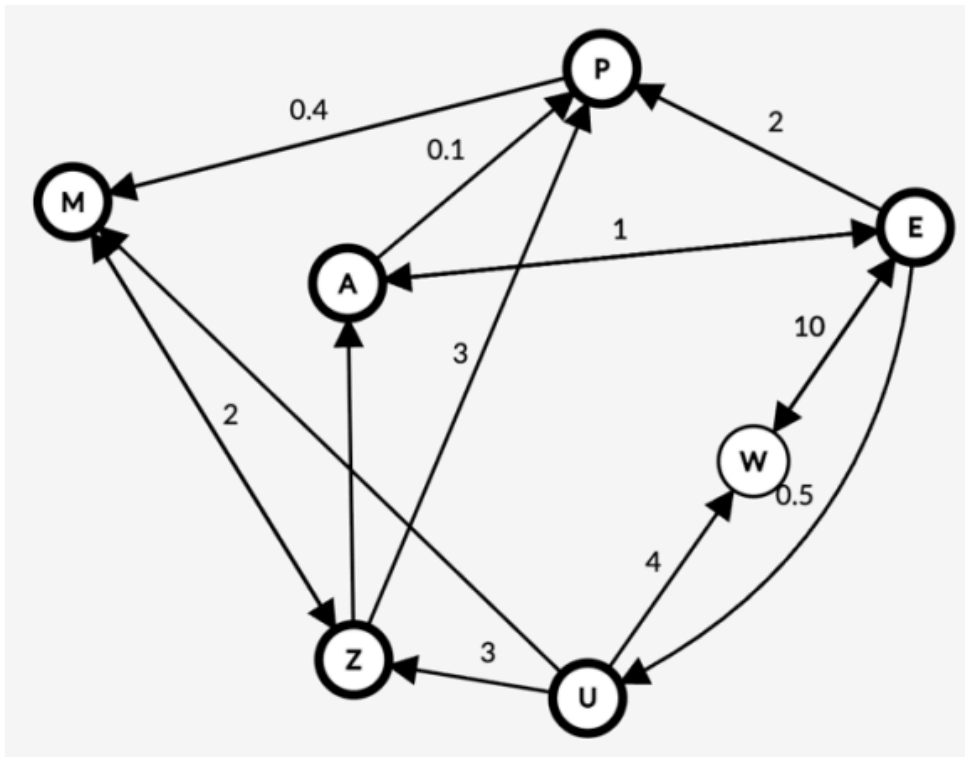


Redo Assignment Test Problem: Dijkstra's

Given the graph below, find the shortest path from Z to all other nodes using Dijkstra's algorithm by hand (Show ALL your work and intermediate steps)

*** NOTE: when the edges don't have a weight I am assuming it is 0 instead of 1

1. Make a table where you have all the nodes and you set the distance from the start to infinity and then the distance from start to itself is just the 0
2. You also need a previous list and visited list
3. Pick the node from the table with the smallest distance and then since the only one with a distance right now is the start of 0 you pick that one and mark it as visited
4. Compute every neighbor of Z where new distance is the distance from Z plus the edge weight
 - a. Only update it for the list if it is smaller than the current one in there
 - b. Update parent pointer as well
 - c. \mark node as visited



① Dijkstra's by hand

Nodes	distance (z)	parents
Z	0	null
A	∞	
U	∞	
M	∞	
W	∞	
E	∞	
P	∞	

② initialize, set all distances ✓

all parents = null

All unvisited

visited

0 = unvisited
1 = visited

Dist = parent distance + edge weight from par → curr

③ Pick smallest distance to visit 1st = Z of 0

see if dist can be updated
update visited
update parents

Node	distance	parents	visited
Z	Dist = 0 + 0 = 0 = cannot be updated still [0]	-1/null	1

Directed so I can only really reach
Z → A → P → M ??
No A → Z

④ Now I check Z's neighbors (edge going out of Z) = A only

Dist = Z⁽⁰⁾ + edge (0) = 0
parent update + visited

Node	Dist	parent	visited
Z	0	null	1
A	0	Z	1
P	∞	null	0
E	∞	null	0

⑤ A only has neighbors = P

A → P = 0.1
= 0 + 0.1 = 0.1

Node	Dist	parent	visited
Z	0	null	1
A	0	Z	1
P	0.1	A	1
E	1	A	0

explore A neighbors = P + E

E = 0 + 1 = 1

visit P bc smallest distance

⑤ only going to M

Node	Dist	parent	visited
Z	0	null	1
A	0	Z	1
E	1	A	0
P	0.1	A	1
M	0.5	P	1

$$= 0.1 + 0.4 = 0.5$$

Explore M next bc smallest distance, mark as visited but no neighbors so chose next smallest dist. unvisited which = E

⑥ ~~No outgoing edges from M - break~~

visit E

Node	Dist	par	visited
E	1	A	1
P	0.1	A	1
W	11	E	0
U	1.5	E	0
A	0	Z	1

= outgoing to P, W, U, A

$$E \rightarrow P = 1 + \text{edge}(2) = 3$$

- larger so no update

$$E \rightarrow W = 1 + \text{edge}(10) = 11$$

$$E \rightarrow U = 1 + 0.5 = 1.5 \text{ update}$$

$$A \rightarrow E = 1 + 1 = 2, \text{ larger}$$

no update

⑦ smallest unvisited = U at 1.5, neighbors = Z, M, W

Node	Dist	par	visited
U	1.5	E	1
Z	0	null	1
M	0.5	P	1
W	5.5	U	0

$$Z = 1.5 + 3 = 3.5 = \text{large no update}$$

$$M = 1.5 + 0 = 1.5 = \text{larger}$$

no update

$$W = 1.5 + 4 = 5.5 = \text{update}$$

8. CURRENT CHART (parent distances)

node	dist	par	visited
Z	0	null	1
M	0.5	P	1
U	1.5	E	1
W	5.5	U	1
A	0	Z	1
P	0.1	A	1
E	1	A	1

Now I have visited all, now I find shortest path from Z-end
now I do reconstruction

smallest unvisited = W

~~no outgoing edges - done~~

neighbor = E

W 5.5 U 1

$w_e = 5.5 + 10 = 15.5$ no update

E 1 A 1

chart for Z paths (path reconstruction)

Z	0	Z
A	0	Z → A
P	0.1	Z, A, P!
M	0.5	Z, A, P, M
U	1.5	Z, A, E, U
E	1	Z A E
W	5.5	Z A E U W

1. start at destination

2. walk backward by looking at parent and going to that node until you reach start

3. cumulatively add distances as you go